DentAI : AI-Powered Dental Assessment

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***Abstract*—The rapid advancement of artificial intelligence (AI) in healthcare offers new opportunities to improve early disease detection and accessibility. This study presents DentAI, a web-based application designed to assist in the preliminary assessment of dental conditions and streamline the appointment booking process. The project aims to address common issues such as delayed diagnosis, dental anxiety, and the lack of accessible self-screening tools. The system utilizes a YOLOv5 deep learning model to detect conditions like gingivitis, ulcers, caries, and discoloration from intraoral images. Built with React (frontend), Flask (backend), and Firebase (database), the platform allows patients to upload images for instant analysis and book appointments with clinics. User Acceptance Testing (UAT) results indicate that the system is effective, user-friendly, and capable of functioning as a reliable decision-support tool for oral health.**

***Index Terms*—Artificial Intelligence, Dental Assessment, YOLOv5, Teledentistry, Web Application, Healthcare**

I. Introduction

Oral health is a critical component of overall well-being, yet it is often neglected due to a lack of awareness and barriers to accessing professional care. Recent technological developments have paved the way for teledentistry and AI-driven diagnostics, which can bridge the gap between patients and dental practitioners [1]. However, many existing solutions are fragmented, offering either scheduling features or diagnostic tools, but rarely both in a unified platform.

This study introduces **DentAI**, an integrated web platform that leverages computer vision to provide automated dental assessments. By combining AI analysis with a clinic management system, DentAI enables users to perform preliminary self-checks and seamlessly proceed to professional treatment. The system is designed to promote proactive oral healthcare, reduce the fear associated with dental visits, and improve operational efficiency for dental clinics.

II. Project Background

Dental diseases such as caries and periodontal disease are prevalent globally but are largely preventable with early detection. Traditional dental care relies heavily on physical visits, which can be hindered by geographic isolation, high costs, or psychological barriers like dental anxiety. Small and medium-sized dental clinics also face challenges in managing appointments and patient records efficiently without modern digital tools.

To overcome these drawbacks, DentAI was developed using the **Rapid Application Development (RAD)** methodology. It incorporates a trained **YOLOv5** model for object detection, a **Flask** API for real-time inference, and a **React** interface for user interaction. This project aims to provide a scalable, cost-effective solution that empowers individuals to monitor their oral health while assisting clinics in digitizing their operations.

III. Problem Statement

A. Accessibility and Delayed Diagnosis

Many dental conditions develop without noticeable symptoms in their early stages. Populations in rural or underserved areas often lack immediate access to dental professionals, leading to delayed diagnoses [2]. Consequently, minor issues often progress into severe complications requiring complex and costly treatments before they are identified.

B. Fear and Delay in Treatment

Dental anxiety is a significant barrier to oral healthcare, with many patients avoiding routine checkups due to fear of procedures or judgment [3]. This avoidance behavior exacerbates oral health issues, as patients often wait until pain becomes unbearable to seek help [4]. There is a lack of non-invasive, private self-assessment tools that could encourage these individuals to seek care earlier.

C. Fragmentation in Health Tech Solutions

Current digital health technologies in dentistry are often siloed. Patients must use separate platforms for telehealth consultations, appointment booking, and educational resources. This fragmentation creates a disjointed user experience, where a patient might receive a digital diagnosis but lacks a direct, automated pathway to schedule a follow-up appointment with a local clinic.

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IV. Research Objectives

The objectives of this project are:

1) To study existing dental-related applications and object detection models suitable for analysing intraoral dental images.
2) To develop a web based Dental assessment platform that utilizes a trained AI model to perform dental condition assessment
3) To evaluate the effectiveness and the accuracy of the DentAI platform by going through user acceptance testing.

V. Literature Review

A. AI in Dental Diagnosis

Artificial Intelligence, particularly Convolutional Neural Networks (CNNs), has transformed medical imaging. Research by Makarim et al. [5] and Zhang et al. [6] highlights the efficacy of object detection models like **YOLOv5** in identifying dental conditions such as caries and gingivitis from intraoral photos. These models offer real-time processing capabilities superior to traditional classification methods, making them suitable for web-based consumer applications.

B. Web Applications in Healthcare

The integration of AI into web platforms requires a robust architecture. The stack comprising **React** (Frontend), **Flask** (AI Backend), and **Firebase** (Database) has been validated in recent studies [7] for its modularity and security. This architecture supports the real-time data flow required for instant image analysis and secure handling of patient health records.

C. Gap Analysis of Existing Solutions

Existing platforms like *Toothpic* or government clinic systems often suffer from limitations. Commercial apps may focus solely on cosmetic consultations or lack instant AI feedback, relying instead on asynchronous review by dentists. Conversely, traditional clinic systems focus on booking but lack diagnostic features. DentAI bridges this gap by offering immediate AI-based screening followed by direct clinical integration [8].

D. Case Studies

Existing dental healthcare platforms and traditional clinical models were studied to gather insights into current service limitations and feature gaps. This information serves as the foundation for the development of **DentAI**, identifying key areas where artificial intelligence can enhance accessibility and operational efficiency compared to current solutions.

1) **Government Dental Clinics:** This represents the traditional public healthcare sector, which provides affordable dental services but often relies on manual operations. These clinics typically use phone-based booking systems and physical record-keeping, leading to long waiting times and limited accessibility outside of office hours. The DentAI project contrasts with this model by digitizing the entire process; unlike government clinics where diagnosis only happens during a physical visit, this system uses AI to provide a preliminary assessment before the patient leaves home. This reduces unnecessary visits and streamlines the flow of patients compared to the fragmented manual process used in public clinics.
2) **Toothpick.com:** This is a commercial teledentistry platform that connects patients with dentists for remote consultations. It focuses on allowing users to upload photos for professional review, acting primarily as a marketplace for dental advice. While it offers a digital solution similar to this project, it relies on asynchronous human review, meaning patients must wait for a dentist to manually check their images. DentAI distinguishes itself by integrating a custom **YOLOv5 AI model** that provides instant, real-time feedback without human delay. Additionally, while Toothpick focuses on the consultation, the proposed system creates a complete end-to-end loop by combining immediate self-assessment with direct appointment scheduling and progress tracking.

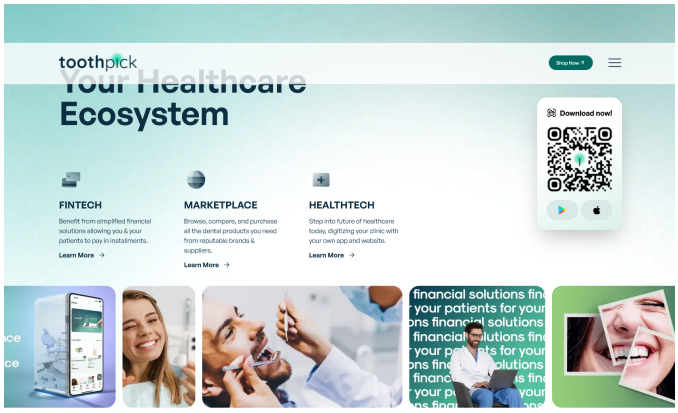


Fig. 1. Toothpick.com

VI. Methodology

A. Rapid Application Development (RAD) Model

The Rapid Application Development (RAD) methodology was selected for the development of DentAI due to its emphasis on iterative prototyping and flexibility. Unlike linear models such as Waterfall, RAD allows for continuous user feedback and rapid adjustments, which is critical when integrating experimental components like AI models.

The development process was divided into four key phases:

1) **Requirements Planning:** Initial stakeholders (supervisors and target users) were consulted to define the core requirements: a web-based platform capable of detecting four specific dental conditions (gingivitis, ulcers, caries, discoloration) and a clinic management system.
2) **User Design (Prototyping):** This phase involved multiple iterations. The system was developed in modular blocks—first the user interface (React), followed by the backend API (Flask). Crucially, the AI model

(YOLOv5) underwent several training cycles (Prototype 1, Prototype 2) to improve detection accuracy based on validation results.

3) **Construction:** The validated prototypes were converted into the final production-ready system. This phase focused on integrating the frontend with the Firebase database and ensuring the Flask API could handle real-time image requests efficiently.
4) **Cutover:** The final phase involved deployment and User Acceptance Testing (UAT). The system was tested by 32 respondents to validate its usability and functionality before the final release.

Overall, the RAD model's iterative nature ensured that issues with the AI's accuracy were identified and resolved early, resulting in a more robust final solution.

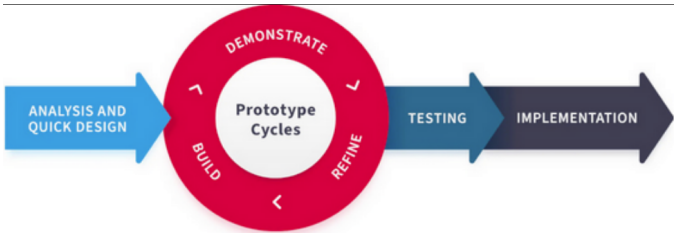


Fig. 2. Rapid Application Development Model (RAD)

VII. System Architecture

The system architecture follows a three-tier design to ensure separation of concerns and scalability:

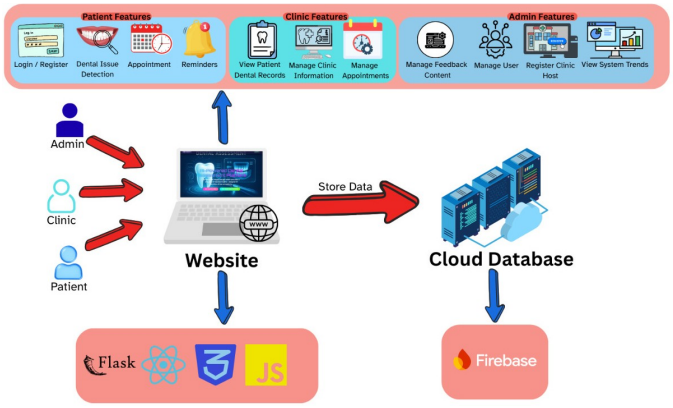


Fig. 3. System Architecture

- **Presentation Layer (Frontend):** Developed using **React.js**, providing a responsive interface for Patients (to upload images/book) and Admins (to manage dashboards).
- **Application Layer (Backend & AI):** Built with **Flask** (Python). This layer hosts the **YOLOv5** model, processing images received from the frontend and returning diagnostic predictions (bounding boxes and labels).
- **Data Layer (Database): Google Firebase** is used for authentication and storing real-time data, including user profiles, appointment schedules, and dental history records.

VIII. Prototype and Use Cases

The system supports three primary user roles, as defined in the system design:

1) System Administrator: The System Administrator has several interactions with the system:

- The system allows the administrator to login and access the Admin dashboard.
- The system allows the administrator to view a summary of total users and clinics.
- The system allows the administrator to view disease detection trends and analytics.
- The system allows the administrator to view the list of registered patients.
- The system allows the administrator to manage patient accounts (view/delete).
- The system allows the administrator to view the list of registered clinics.
- The system allows the administrator to add clinic accounts.
- The system allows the administrator to manage feedback content.
- The system allows the administrator to logout.

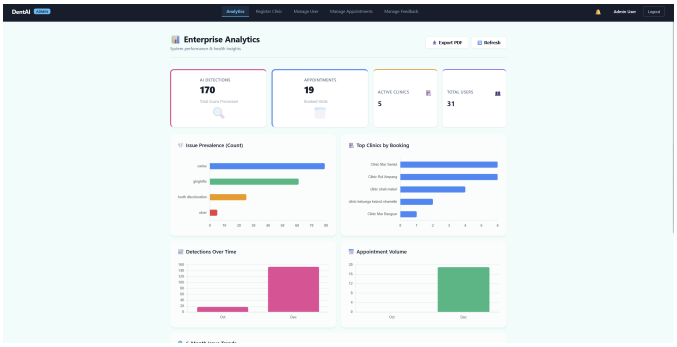


Fig. 4. admin dashboard

2) Clinic Admin: The Clinic Admin interacts with the system in the following ways:

- The system allows the clinic admin to login and access the clinic panel.
- The system allows the clinic admin to view the appointment dashboard.
- The system allows the clinic admin to view incoming appointment requests.
- The system allows the clinic admin to manage appointment bookings.
- The system allows the clinic admin to view patient details and medical history.
- The system allows the clinic admin to view AI analysis results associated with a patient.
- The system allows the clinic admin to update clinic profile information.
- The system allows the clinic admin to logout.

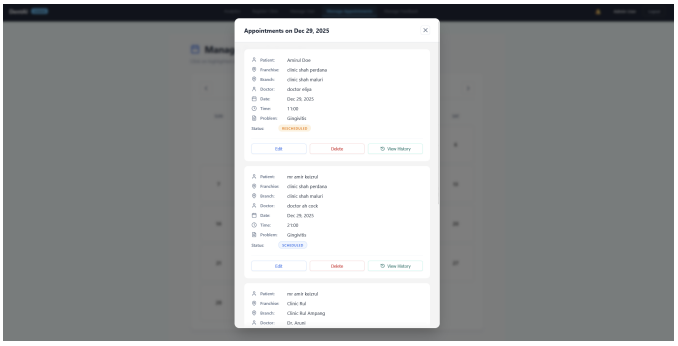


Fig. 5. Clinic admin manage appointment

3) Patient: The Patient interacts with the system as follows:

- The system allows the patient to register a new account.
- The system allows the patient to login and access the landing page.
- The system allows the patient to upload intraoral images for assessment.
- The system allows the patient to view AI-generated analysis results (e.g., Gingivitis, Caries).
- The system allows the patient to view a history of past assessments.
- The system allows the patient to browse a list of available dental clinics.
- The system allows the patient to select a doctor and view available time slots.
- The system allows the patient to book an appointment.
- The system allows the patient to view and manage their bookings.
- The system allows the patient to manage their personal profile.
- The system allows the patient to logout.

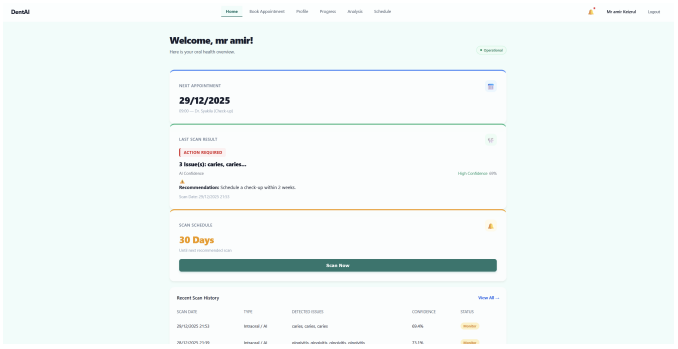


Fig. 6. Patient Dashboard

IX. Conclusion and Recommendations

The **DentAI** system successfully integrates artificial intelligence with dental practice management, providing a novel solution for early self-assessment. The system acts as a decision-support tool, empowering patients to take proactive steps toward their oral health while streamlining clinic operations.

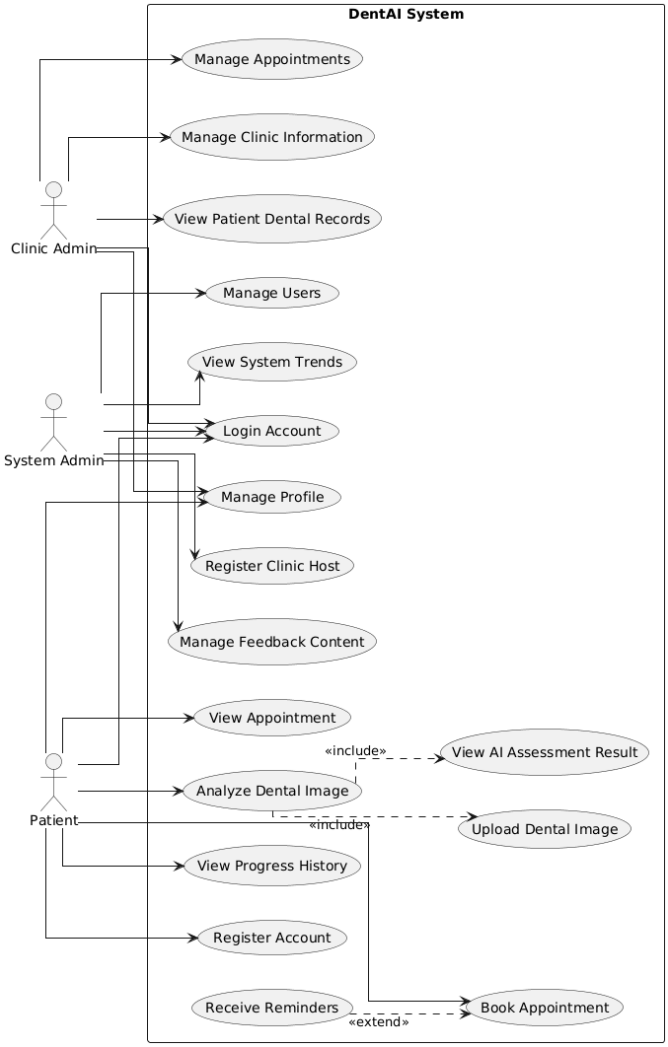


Fig. 7. Use Case Diagram for DentAI:AI-Powered Dental Assessment

UAT results confirm that users find the flow from AI diagnosis to appointment booking logical and valuable.

X. Future Works

- **Future Work 1:** Additional dental disease classes can be included in the AI model,such as periodontitis, plaque buildup, tooth fractures, and abscesses. Expanding the number of detectable conditions would allow the system to provide more comprehensive dental assessments.
- **Future Work 2:** The model can be trained using larger and more diverse datasets collected from multiple dental clinics. A bigger dataset with varied lighting conditions, image angles, and patient demographics can help improve model accuracy, reduce prediction errors, and enhance generalization.
- **Future Work 3:**The overall system can be enhanced by improving user interface design, system performance, and scalability. Future work may include better appointment management features, real-time notification systems, and

cloud optimization to support a larger number of users and clinics efficiently.

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